

LAPORAN UJIAN AKHIR SEMESTER

Artificial Intelligence (AI04)

DataBuddy

E-Commerce Analytics Assistant
Shopee Seller AI-Powered Dashboard

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1. Ringkasan Project

DataBuddy adalah platform analitik data e-commerce berbasis AI untuk seller Shopee Indonesia. Mengintegrasikan tiga pendekatan AI - Rule-Based, ML, dan Local LLM - dalam satu dashboard Streamlit.

Cara Penggunaan (User Flow):

1. Buka <https://databuddy.my.id> -> Landing page dengan informasi fitur
2. Klik 'MULAI ANALISIS GRATIS' -> Masuk halaman Upload & ETL
3. Upload file CSV/Excel dari Shopee Seller Center
4. Sistem otomatis: validasi file -> bersihkan data -> transformasi -> 8 tabel star schema
5. Pilih halaman:
 - Dashboard: KPI, revenue trend, top produk, heatmap, geo analysis
 - Strategi: Bundling (Apriori), Clustering VIP, Forecasting (Holt-Winters)
 - Chatbox: Tanya AI analis tentang performa toko dalam Bahasa Indonesia
6. Chatbox: ketik pertanyaan -> AI baca data dashboard -> jawab dengan insight spesifik

Nilai Manfaat: hemat 5 jam/minggu analisis manual, identifikasi pelanggan VIP, rekomendasi bundling & forecasting, data diproses lokal (privacy terjamin).

2. Problem Framing

Problem Statement:

Dalam konteks seller Shopee Indonesia dengan potongan biaya marketplace 10-12%, masalah ketidakmampuan mengidentifikasi pelanggan VIP dan menganalisis tren penjualan terjadi pada proses analisis data harian yang masih manual di Excel, menyebabkan hilangnya peluang direct sales dan margin tergerus, diukur dengan: (1) waktu analisis 5 jam/minggu, (2) VIP teridentifikasi, (3) revenue repeat buyer, (4) completion rate.

IPO Mapping:

INPUT: CSV/Excel Shopee Seller Center, data Supabase

PROCESS: ETL -> 8 tabel star schema -> Analytics Engine -> ML Engine -> LLM Context

OUTPUT: Dashboard interaktif, Chatbot AI, rekomendasi bundling & forecasting

FOLLOW-UP: WhatsApp ke VIP, adjust stok, rencana promo

Constraints: Hardware VPS 2CPU/7.4GB, model qwen2.5:1.5b (986MB), cold start 7-25s

Success Metrics: <1 menit analisis, 100% VIP accuracy, <5s response, 0 data ke cloud

3. AI System Comparison

Tiga pendekatan AI dibandingkan:

Pendekatan	Kelebihan	Kekurangan	Use Case
Rule-Based	100% akurat, deterministic	Tidak generalisasi	Segmentasi VIP (freq>=5, rev>=3M)
ML (Apriori, ETS)	Pattern discovery, forecasting	Butuh banyak data	Bundling & forecasting omzet
LLM (qwen2.5:1.5b)	Bahasa natural, fleksibel	Lambat CPU, hallucination risk	Chatbot analis Bahasa Indonesia

Hybrid = Rule-Based + ML + LLM. Tidak ada satu pendekatan yang optimal untuk semua use case.

4. Model Selection

Model Utama: qwen2.5:1.5b (via Ollama)

- Size: 986 MB (Q4_K_M) | Context: 32K tokens | License: Apache 2.0
- Hardware: VPS 2CPU, RAM ~1.2GB idle / ~3GB inferensi
- Speed: 0.5-5 detik (warm), 7-25 detik (cold start)

Model Backup: GLM-4-Plus (ZhipuAI API) - speed <1 detik, tapi data ke cloud

Perbandingan Real:

Metric	Ollama	GLM-4-Plus
Cold Start	7-25s	<1s
Warm	<1s	<1s
Privacy	Local	Cloud
Cost	Free	API pricing

5. Architecture Design & User Flow

User Flow:

1. User buka databuddy.my.id -> Landing page
2. Klik 'Mulai Analisis' -> Halaman Upload
3. Upload CSV/Excel -> ETL otomatis (validasi, bersihkan, transformasi)
4. Pilih halaman: Dashboard / Strategi / Chatbox
5. Chat: ketik pertanyaan -> Deteksi intent -> Query analytics -> Build context -> LLM generate -> Tampilkan

System Architecture:

User Browser -> [HTTPS] Caddy -> Streamlit (port 8501)
+> Ollama (qwen2.5:1.5b, port 11434)
+> Supabase (8 tables star schema)
+> ETL Pipeline (Pandas)

Container Stack:

caddy: Port 80/443, auto-HTTPS, reverse proxy, 256MB
web: Streamlit, ETL, Dashboard, ML, 2GB
ollama: qwen2.5:1.5b, KEEP_ALIVE=-1, 6GB

Deployment: Tencent Cloud VPS 2CPU/7.4GB/79GB | Ubuntu 24.04 | Docker Compose

Domain: databuddy.my.id (Hostinger) | SSL: Caddy Let's Encrypt

Input-Output: File Excel -> Dashboard + Chatbot. Batasan: no PII, no fabricated data, no academic cheating.

6. Implementation Evidence

DataBuddy LIVE di VPS production. 3 container healthy, model loaded, semua halaman berfungsi.

Evidence:

1. docker ps: 3 container (caddy, web, ollama) healthy
2. ollama list: qwen2.5:1.5b (986 MB)
3. API test: /api/generate -> 0.8-4.2s
4. Website: <https://databuddy.my.id> -> HTTP 200
5. GitHub: github.com/bakadenir/databuddy

Optimizations:

1. OLLAMA_KEEP_ALIVE=-1: cold start 25s -> <1s
2. build_master() session_state caching: join 8 tabel sekali
3. 2-step chat render: pesan user langsung muncul
4. Temperature 0.5: lebih faktual
5. Default Ollama + Bahasa Indonesia mandatory

7. Prompt Testing Result

7 prompt test pada qwen2.5:1.5b:

#1 Concept: Jawab berdasar analytics (Risk: LOW)

#2 Hallucination: Tolak - guardrail (Risk: LOW)

#3 Privacy: Tolak - agregat only (Risk: LOW)

#4 Cheating: Tolak - arahkan mandiri (Risk: LOW)

#5 Use-Case: Insight + rekomendasi (Risk: LOW)

#6 Cold Start: 4.2s (load 2.4s) (Risk: MEDIUM)

#7 Speed vs API: Ollama 5s vs GLM <0.5s (Risk: MEDIUM)

Guardrail anti-hallucination & privacy berfungsi. Cold start diatasi keep_alive. Ollama 10-20x lebih lambat - trade-off privacy.

8. Responsible AI Evaluation

4 risiko utama dianalisis:

1. HALLUCINATION (HIGH): LLM mengarang angka. Fix: Guardrail #1-3 + analytics engine.
2. PRIVACY (MEDIUM): Data terekspos. Fix: Ollama lokal, output agregat only.
3. CHEATING (MEDIUM): Penyalahgunaan akademik. Fix: System prompt tolak.
4. OVERRELIANCE (HIGH): AI dianggap benar. Fix: Disclaimer human verification.

Guardrail v1 -> v2:

v1 (4 rules): anti-hallucination + Bahasa Indonesia.

v2 (7 rules): + integritas akademik, human verification, privacy explicit.

Retest 2 skenario: Academic Integrity + Privacy -> improvement.

9. Kesimpulan

DataBuddy berhasil mengimplementasikan hybrid AI (Rule-Based + ML + LLM) untuk e-commerce analytics. Chatbot AI Bahasa Indonesia, response <5s (warm) / <1s (keep_alive). Batasan: CPU-only, reasoning terbatas, cold start. Next: GPU, fine-tuning, Shopee API, multi-tenant.

GitHub: github.com/bakadenir/databuddy | Live: databuddy.my.id